

Overview of RMSprop optimizer

Subject: RMSprop optimizer

1.

This procedure will remain numerically stable virtually for all η as the learning rate is now normalized. Such comparison between classical and implicit stochastic gradient descent in the least squares problem is very similar to the comparison between least mean squares (LMS) and normalized least mean squares filter (NLMS). Even though a closed-form solution for ISGD is only possible in least squares, the procedure can be efficiently implemented in a wide range of models. Specifically, suppose that $Q_i(w)$ depends on w only through a linear combination with features x_i , so that we can write $\nabla_w Q_i(w) = -q(x_i'w)x_i$, where $q(\cdot) \in \mathbb{R}$ may depend on x_i, y_i as well but not on w except through $x_i'w$. Least squares obeys this rule, and so does logistic regression, and most generalized linear models. For instance, in least squares, $q(x_i'w) = y_i - x_i'w$, and in logistic regression $q(x_i'w) = y_i - S(x_i'w)$, where $S(u) = e^u / (1 + e^u)$ is the logistic function. In Poisson regression, $q(x_i'w) = y_i - e^{x_i'w}$, and so on. In such settings, ISGD is simply implemented as follows. Let $f(\xi) = \eta q(x_i'w_{\text{old}}) + \xi \|x_i\|^2$, where ξ is scalar. Then, ISGD is equivalent to:

2.

$G_{j,j} = \sum_{\tau=1}^t g_{j,\tau}^2$. This vector essentially stores a historical sum of gradient squares by dimension and is updated after every iteration. The formula for an update is now

3.

where, γ is the forgetting factor. The concept of storing the historical gradient as sum of squares is borrowed from Adagrad, but "forgetting" is introduced to solve Adagrad's diminishing learning rates in non-convex problems by gradually decreasing the influence of old data. And the parameters are updated as,

4.

$$dW_t = -\nabla(Q(W_t) + \frac{1}{4\eta} \|\nabla Q(W_t)\|^2) dt + \eta \Sigma(W_t)^{1/2} dB_t, \\ dW_t = -\nabla \left(Q(W_t) + \frac{1}{4\eta} \|\nabla Q(W_t)\|^2 \right) dt + \sqrt{\eta} \Sigma(W_t)^{1/2} dB_t,$$

5.

$$v(w, t) := \gamma v(w, t-1) + (1-\gamma) \|\nabla Q_i(w)\|^2 \\ v(w, t-1) + \left(1-\gamma\right) \|\nabla Q_i(w)\|^2$$

6.

where E denotes taking the expectation with respect to the random choice of indices in the stochastic gradient descent scheme. Since this approximation does not capture the random fluctuations around the mean behavior of stochastic gradient descent solutions to stochastic differential equations (SDEs) have been proposed as limiting objects. More precisely, the solution to the SDE

7.

Linear regression Suppose we want to fit a straight line $\hat{y} = w_1 + w_2 x$ to a training set with observations $((x_1, y_1), (x_2, y_2), \dots, (x_n, y_n))$ and corresponding estimated responses $(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$ using least squares. The objective function to be minimized is

8.

Implicit updates (ISGD) As mentioned earlier, classical stochastic gradient descent is generally sensitive to learning rate η . Fast convergence requires large learning rates but this may induce numerical instability. The problem can be largely solved by considering implicit updates whereby the stochastic gradient is evaluated at the next iterate rather than the current one:

9.

Adam Adam (short for Adaptive Moment Estimation) is a 2014 update to the RMSProp optimizer combining it with the main feature of the Momentum method. In this optimization algorithm, running averages with exponential forgetting of both the gradients and the second moments of the gradients are used. Given parameters $w(t)$ and a loss function $L(t)$, where t indexes the current training iteration (indexed at 1), Adam's parameter update is given by:

10.

$\max_{k=0, \dots, \lfloor T/\eta \rfloor} |E[g(w_k)] - g(W_{k\eta})| \leq C\eta$

11.

$w_{\text{new}} := \arg \min_w \{Q_i(w) + \frac{1}{2\eta} \|w - w_{\text{old}}\|^2\}$

12.

History In 1951, Herbert Robbins and Sutton Monro introduced the earliest stochastic approximation methods, preceding stochastic gradient descent. Building on this work one year later, Jack Kiefer and Jacob Wolfowitz published an optimization algorithm very close to stochastic gradient descent, using central differences as an approximation of the gradient. Later in the 1950s, Frank Rosenblatt used SGD to optimize his perceptron model, demonstrating the first applicability of stochastic gradient descent to neural networks. Backpropagation was first described in 1986, with stochastic gradient descent being used to efficiently optimize parameters across neural networks with multiple hidden layers. Soon after,

another improvement was developed: mini-batch gradient descent, where small batches of data are substituted for single samples. In 1997, the practical performance benefits from vectorization achievable with such small batches were first explored, paving the way for efficient optimization in machine learning. As of 2023, this mini-batch approach remains the norm for training neural networks, balancing the benefits of stochastic gradient descent with gradient descent. By the 1980s, momentum had already been introduced, and was added to SGD optimization techniques in 1986. However, these optimization techniques assumed constant hyperparameters, i.e. a fixed learning rate and momentum parameter. In the 2010s, adaptive approaches to applying SGD with a per-parameter learning rate were introduced with AdaGrad (for "Adaptive Gradient") in 2011 and RMSprop (for "Root Mean Square Propagation") in 2012. In 2014, Adam (for "Adaptive Moment Estimation") was published, applying the adaptive approaches of RMSprop to momentum; many improvements and branches of Adam were then developed such as Adadelta, Adagrad, AdamW, and Adamax. Within machine learning, approaches to optimization in 2023 are dominated by Adam-derived optimizers, TensorFlow and PyTorch, by far the most popular machine learning libraries, as of 2023 largely only include Adam-derived optimizers, as well as predecessors to Adam such as RMSprop and classic SGD. PyTorch also partially supports limited-memory BFGS, a line-search method, but only for single-device setups without parameter groups.